5 Two stars A and B are separated by a distance of 1.2×10^{10} m as shown in Fig. 5.1. x is the distance from the centre of star A, in the direction toward the centre of star B.

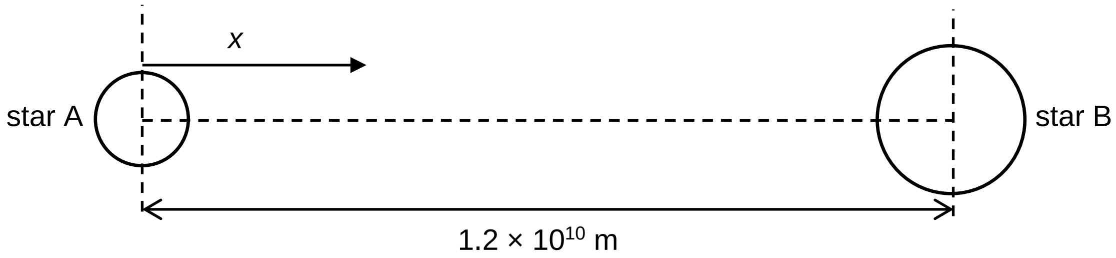

Fig. 5.1

The variation with x of the gravitational potential ϕ due to the two stars along the line joining their centres is shown in Fig. 5.2.

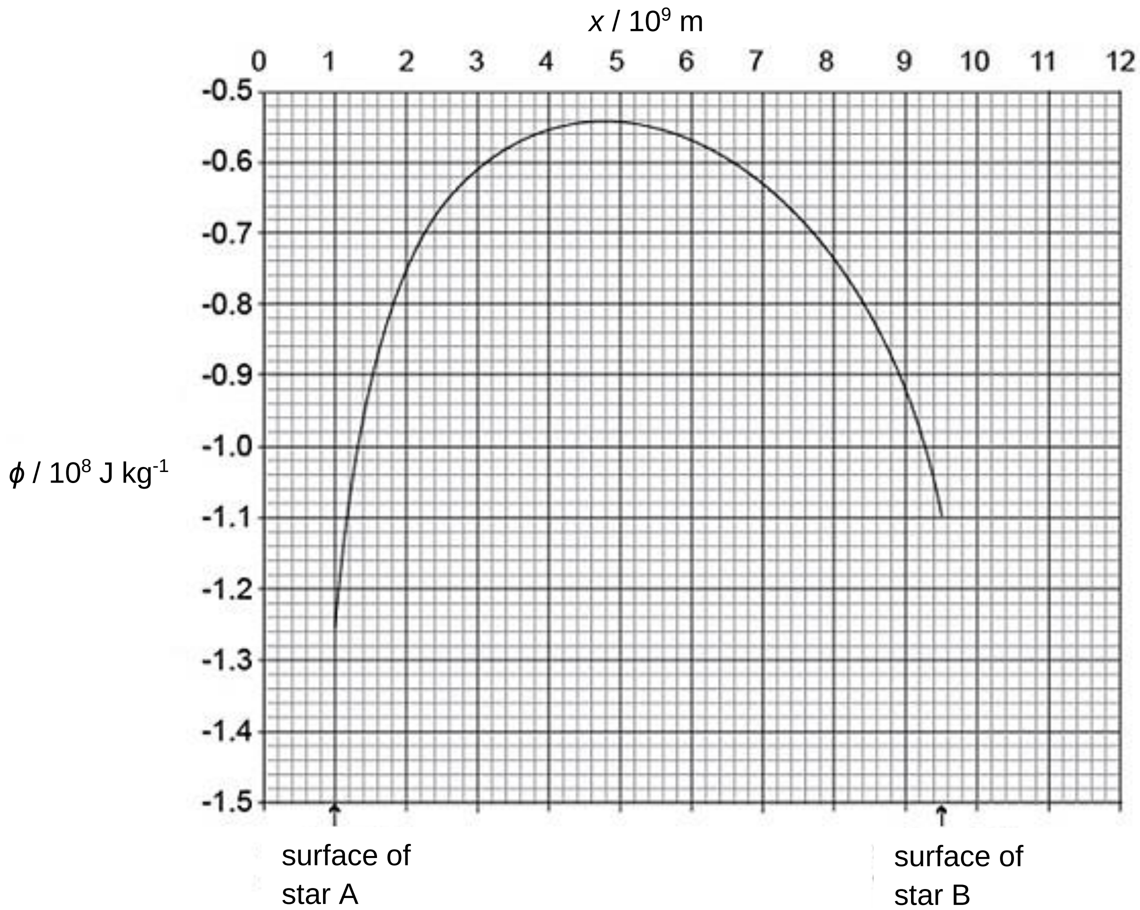

Fig. 5.2

A body is launched with kinetic energy E_K from the surface of star B.

The body then arrives at the surface of the star A.

(a) Define *gravitational potential* at a point.

Gravitational potential at a point in a gravitational field is the work done per unit mass by an external agent in bringing a small test mass from infinity to that point. [B1] [1]

(b) Use Fig. 5.2 to explain whether the kinetic energy of the body when it arrives at the surface of star A is less than, equal to, or larger than E_k.

The potential at the surface of A is smaller than that of B, hence there is a loss in potential energy.[B1] By conservation of energy, the body will have a gain in kinetic energy. Hence the kinetic energy of the body at A is larger than E_k. [B1] [2]

(c) State and explain the distance x at which the resultant gravitational field strength due to the two stars is zero.

The magnitude of the potential gradient is equal to the gravitational field strength, which is zero at $x = 4.8 \times 10^9$ m. [M1]

$x = 4.8 \times 10^9$ m (allow answers in the range of 4.6 to 5.0) [A1] [2]

(d) Determine the ratio $\dfrac{\text{average density of star A}}{\text{average density of star B}}$.

$$\left.\begin{array}{l} \text{At } x = 4.8\times10^9 \text{ m,} \\ \Sigma g = 0 \\ g_A = g_B \end{array}\right\} \text{[M1]}$$

$$\frac{GM_A}{r_A^2} = \frac{GM_B}{r_B^2}$$

$$\frac{G\rho_A \frac{4}{3}\pi R_A^3}{r_A^2} = \frac{G\rho_B \frac{4}{3}\pi R_B^3}{r_B^2}$$

$$\frac{\rho_A}{\rho_B} = \frac{R_B^3 r_A^2}{R_A^3 r_B^2} = \frac{(2.5\times10^9)^3(4.8\times10^9)^2}{(1.0\times10^9)^3[(12.0-4.8)\times10^9]^2} \quad \text{[M1]}$$

$$\frac{\rho_A}{\rho_B} = 6.9 \quad \text{[A1]}$$

Alternatively,

$$\frac{\phi_A}{\phi_B} = \frac{-\frac{GM_A}{r_{AA}} + (-\frac{GM_B}{r_{BA}})}{-\frac{GM_A}{r_{AB}} + (-\frac{GM_B}{r_{BB}})}$$

$$\frac{-1.26\times10^8}{-1.10\times10^8} = \frac{\frac{M_A}{1.0\times10^9} + \frac{M_B}{11.0\times10^9}}{\frac{M_A}{9.5\times10^9} + \frac{M_B}{2.5\times10^9}}$$

$$\frac{M_A}{M_B} = 0.4176$$

$$\frac{\rho_A}{\rho_B} = \frac{M_A}{M_B} \times \frac{V_B}{V_A} = \frac{M_A}{M_B} \times \frac{\frac{4}{3}\pi R_B{}^3}{\frac{4}{3}\pi R_A{}^3}$$

$$= \frac{M_A}{M_B} \times \frac{R_B{}^3}{R_A{}^3} = 0.4176 \times \frac{(2.5\times10^9)^3}{(1.0\times10^9)^3} = 6.5$$

ratio = ... [3]

[Total: 8]

6 A power supply is connected across a load as shown in Fig. 6.1.